

PTVA's MULUND COLLEGE OF COMMERCE (AUTONOMOUS)			
Programme: B.Sc. Information Technology		Class: TY	Semester: V (AUTO)
Exam: ESE (ATKT)	Name of the Course: DevOps		
Date of Exam: 27/11/2025	Marks: 60	Time: 2 hours	Autonomous
Instructions: (1) <u>All</u> questions are <u>compulsory</u> . (2) Make <u>suitable assumptions</u> wherever necessary and <u>state the assumptions</u> made. (3) Answers to the <u>same question</u> must be <u>written together</u> . (4) Numbers to the <u>right</u> indicate <u>marks</u> . (5) Draw <u>neat labeled diagrams</u> wherever <u>necessary</u> . (6) Use of non-programmable calculator is <u>allowed</u> .			

Q 1 Attempt any TWO of the following: 12

- A. Explain in brief the ten practices of continuous integration processes.
- B. How can implementing DevOps practices contribute to enhancing business agility and responsiveness in today's fast-paced market?
- C. Write about how recent technology trends have supported business groups and development teams in bypassing the IT operations shared service completely.
- D. What are some simple strategies to prevent unnecessary motion in parallel development and how can issues be addressed to smooth the flow and reduce wait times?

Q 2 Attempt any TWO of the following: 12

- A. Describe how organizations can assess their API management capabilities to effectively create and integrate APIs and identify the key factors involved in this evaluation.
- B. What new skills will QA and testing professionals need to fully leverage advanced automation as it becomes more pervasive, and how do these skills contribute to providing a comprehensive and elevated test discipline beyond routine tasks?
- C. Discuss the key obstacles to achieving continuous delivery in terms of development challenges and operations challenges and how finding common ground between these areas can help overcome these obstacles.
- D. Which new skills will agile operations teams need to acquire to address the emerging IT operations imperatives and effectively support modern IT environments?

Q 3 Attempt any TWO of the following: 12

- A. In developing a DevOps transformation playbook, why is assessing the current state crucial and how does this step help in effectively planning and implementing the transformation?
- B. How do revenue streams operate for a line of business (LOB) and an IT organization and what is their role in sustaining and driving growth within each of these entities?
- C. To address the emerging IT operations imperatives and effectively support modern IT environments, which new skills will agile operations teams need to acquire?

- D.** From a DevOps plays perspective, what is the role of Continuous Integration as a play and how does it contribute to improving the software development and delivery process?

Q 4 Attempt any TWO of the following:

12

- A.** Describe how leveraging existing reference sources, roles and practices can enhance practical DevOps, particularly in the context of DevOps and Lean Startup methodologies
- B.** What challenges do organizations face when measuring the impact of DevOps on ROI and how are practitioners and industry analysts addressing these challenges to attempt such calculations?
- C.** How can the integration of DevOps and Lean practices improve IT outcomes in the public sector and what benefits does this approach offer for government IT initiatives?
- D.** Why is delivering microservices architectures considered a key DevOps play for driving innovation and how does it contribute to the overall success of DevOps strategies?

Q 5 Attempt any TWO of the following:

12

- A.** How does a DevOps Center of Competency (CoC) contribute to scaling DevOps within an enterprise and in what ways does it support the successful implementation and expansion of DevOps practices across the organization?
- B.** What are the key strategies for scaling DevOps in an enterprise while managing outsourcing relationships and what factors should be considered to ensure successful integration and collaboration with external partners?
- C.** Which strategies can be used to build an environment and culture of trust and how can a formula be proposed to calculate the total time to delivery effectively?
- D.** Discuss how applying DevOps thinking to the line of business can enhance the adoption of DevOps in the enterprise and what benefits this approach offers for aligning business goals with DevOps practices.